

FER2013 — KD to MobileNetV3-Small @ 224

This notebook trains a MobileNetV3-Small student via knowledge distillation from an EfficientNet-B0 teacher. The goal is to retain high accuracy while improving efficiency. Paths and core training code match the earlier version; additional report cells are appended for visualization.

Mounted at /content/drive
 Device: cuda
 CSV: /content/drive/MyDrive/Datasets/fer2013.csv
 Teacher ckpt: /content/drive/MyDrive/ComputerVision_outputs/best_model.pt

Split sizes (train/val/test): 28709 3589 3589

Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_rwightman-7f5810bc.pth

100%|██████████| 20.5M/20.5M [00:00<00:00, 26.2MB/s]

Loaded teacher: efficientnet_b0

Downloading: "https://download.pytorch.org/models/mobilenet_v3_small-047dcff4.pth" to /root/.cache/torch/hub/checkpoints/mobilenet_v3_small-047dcff4.pth

100%|██████████| 9.83M/9.83M [00:00<00:00, 108MB/s]

Student: mobilenet_v3_small

```
/tmp/ipython-input-173403918.py:15: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', args...)` instead.
  scaler = torch.cuda.amp.GradScaler(enabled=True)
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-173403918.py:35: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=True):
/usr/local/lib/python3.12/dist-packages/torch/optim/lr_scheduler.py:192: UserWarning: Detected call of `lr_scheduler.step()` before `optimizer.step()`. In PyTorch 1.1.0 and later, you should call them in the opposite order: `optimizer.step()` before `lr_scheduler.step()`. Failure to do this will result in PyTorch skipping the first value of the learning rate schedule. See more details at https://pytorch.org/docs/stable/optim.html#how-to-adjust-learning-rate
  warnings.warn(
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 1/20] train_loss=2.4177 val_acc=0.3968
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 2/20] train_loss=1.8594 val_acc=0.5344
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 3/20] train_loss=1.5343 val_acc=0.6030
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 4/20] train_loss=1.3455 val_acc=0.6094
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 5/20] train_loss=1.2192 val_acc=0.6222
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 6/20] train_loss=1.1145 val_acc=0.6467
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 7/20] train_loss=1.0264 val_acc=0.6598
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 8/20] train_loss=0.9539 val_acc=0.6626
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 9/20] train_loss=0.8793 val_acc=0.6712
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 10/20] train_loss=0.8189 val_acc=0.6801
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 11/20] train_loss=0.7594 val_acc=0.6801
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 12/20] train_loss=0.7089 val_acc=0.6865
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 13/20] train_loss=0.6636 val_acc=0.6941
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 14/20] train_loss=0.6242 val_acc=0.6955
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 15/20] train_loss=0.6005 val_acc=0.6966
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 16/20] train_loss=0.5725 val_acc=0.7021
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 17/20] train_loss=0.5483 val_acc=0.6980
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 18/20] train_loss=0.5360 val_acc=0.7016
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 19/20] train_loss=0.5271 val_acc=0.7002
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
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/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
[KD 20/20] train_loss=0.5238 val_acc=0.6999
Best student val_acc: 0.7021454444134857
```

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)
  img = Image.fromarray(arr, mode='L')
```

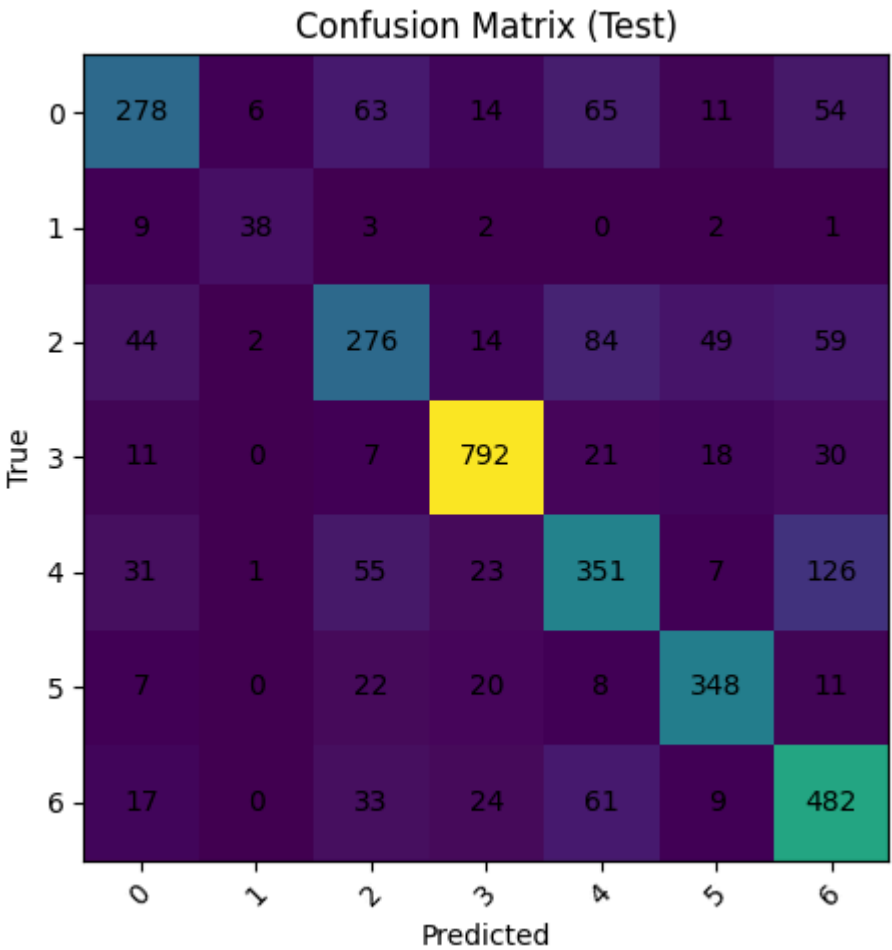


```
TEST accuracy: 0.7146837559208693
      precision    recall  f1-score   support
0         0.700252    0.566191    0.626126     491.000000
1         0.808511    0.690909    0.745098      55.000000
2         0.601307    0.522727    0.559271     528.000000
3         0.890889    0.901024    0.895928     879.000000
4         0.594915    0.590909    0.592905     594.000000
5         0.783784    0.836538    0.809302     416.000000
6         0.631717    0.769968    0.694024     626.000000
accuracy         0.714684    0.714684    0.714684
macro avg        0.715911    0.696895    0.703236    3589.000000
weighted avg     0.714339    0.714684    0.711769    3589.000000
GFLOPs/img (approx Conv+Linear): 0.055
Efficiency score (Accuracy% / GFLOPs): 1287.9
Saved metrics to /content/drive/MyDrive/FER2013_distilled_mnv3small/mnv3small_student_metrics.txt
```

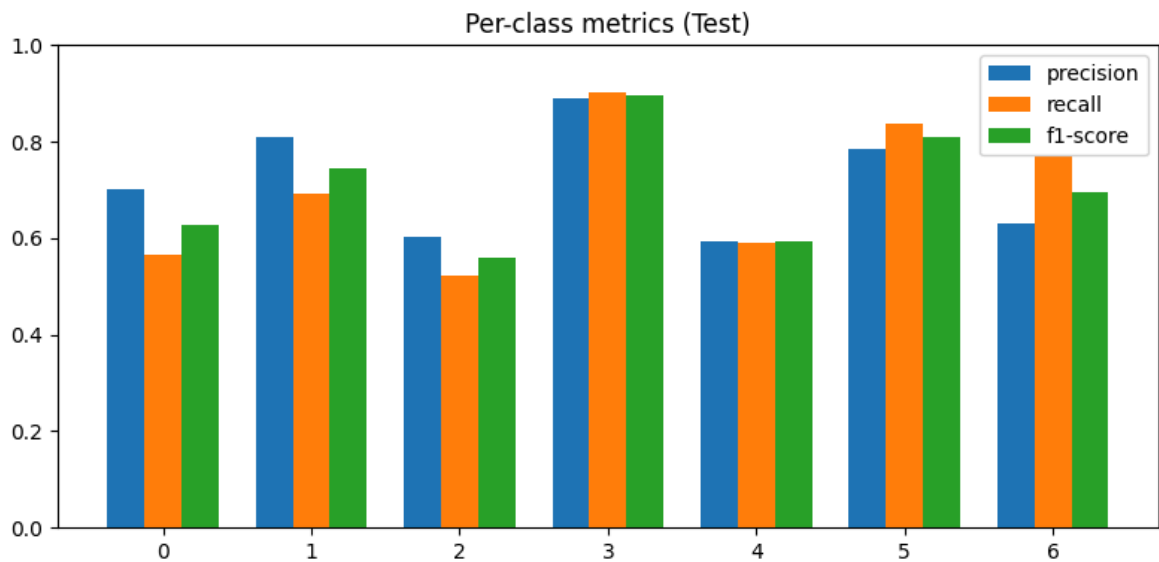
Report: Visual Analysis and Efficiency Summary

This section includes visualizations and summaries based on the results computed above.

Confusion Matrix

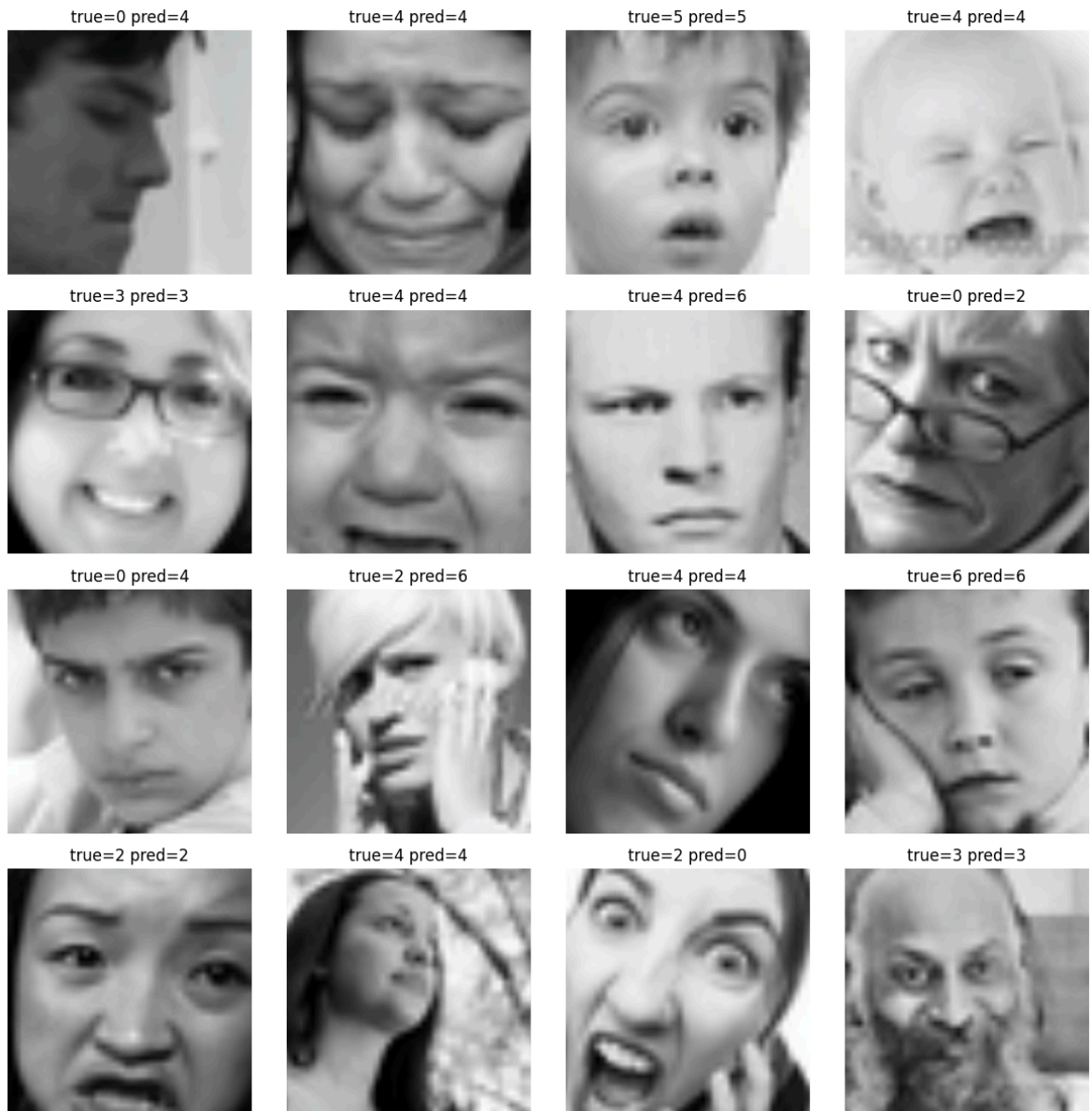


Per-class Precision, Recall, and F1

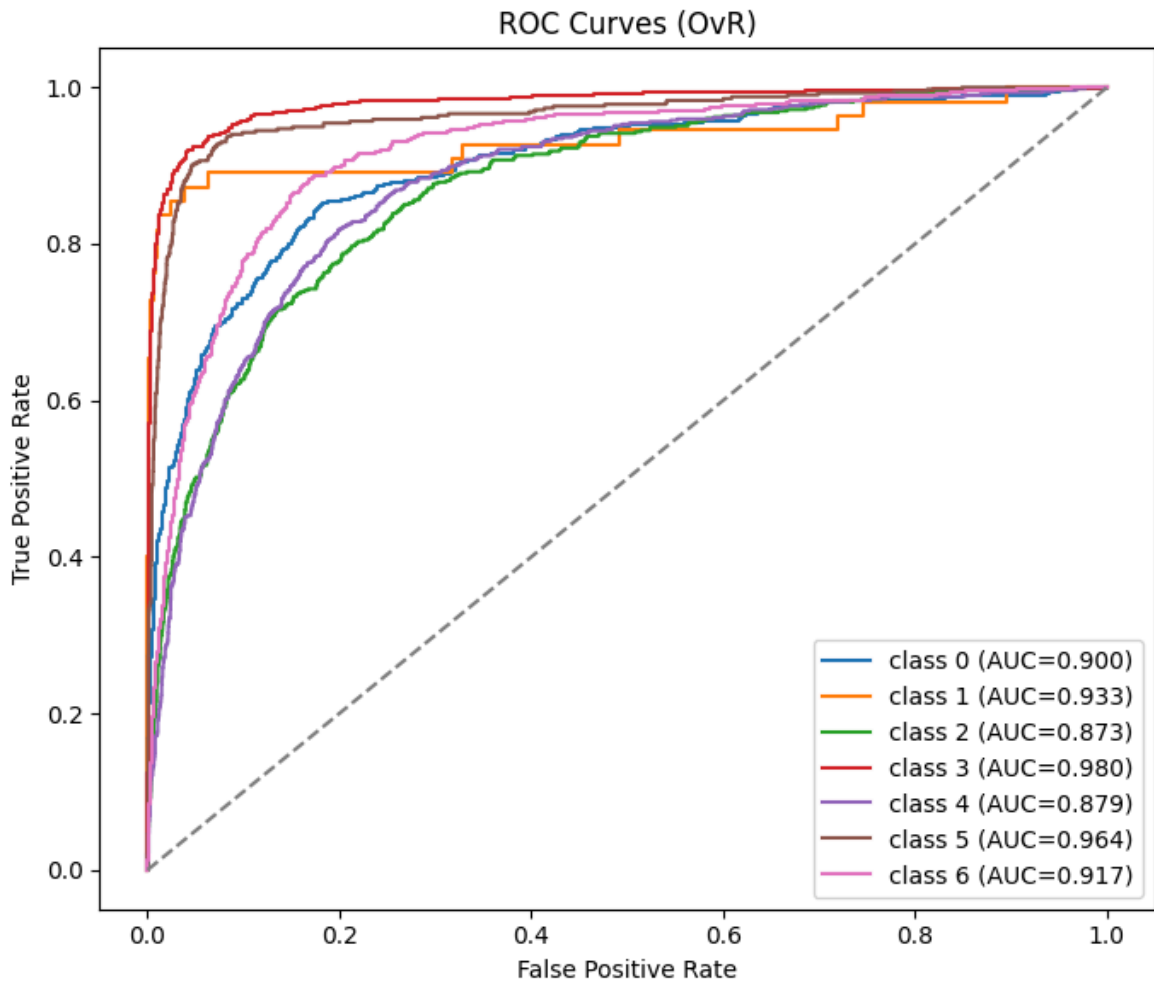


Sample Predictions (Random Test Images)

```
/tmp/ipython-input-2238052287.py:66: DeprecationWarning: 'mode' parameter is deprecated and will be removed in Pillow 13 (2026-10-15)  
img = Image.fromarray(arr, mode='L')
```

ROC Curves (One-vs-Rest)



Per-class AUCs: [0.8999446459774981, 0.9327622575500333, 0.873074505261699, 0.9799894210546202, 0.8789494837074137, 0.9643139076098814, 0.9171420900369737]

Inference Latency and Throughput (Optional)

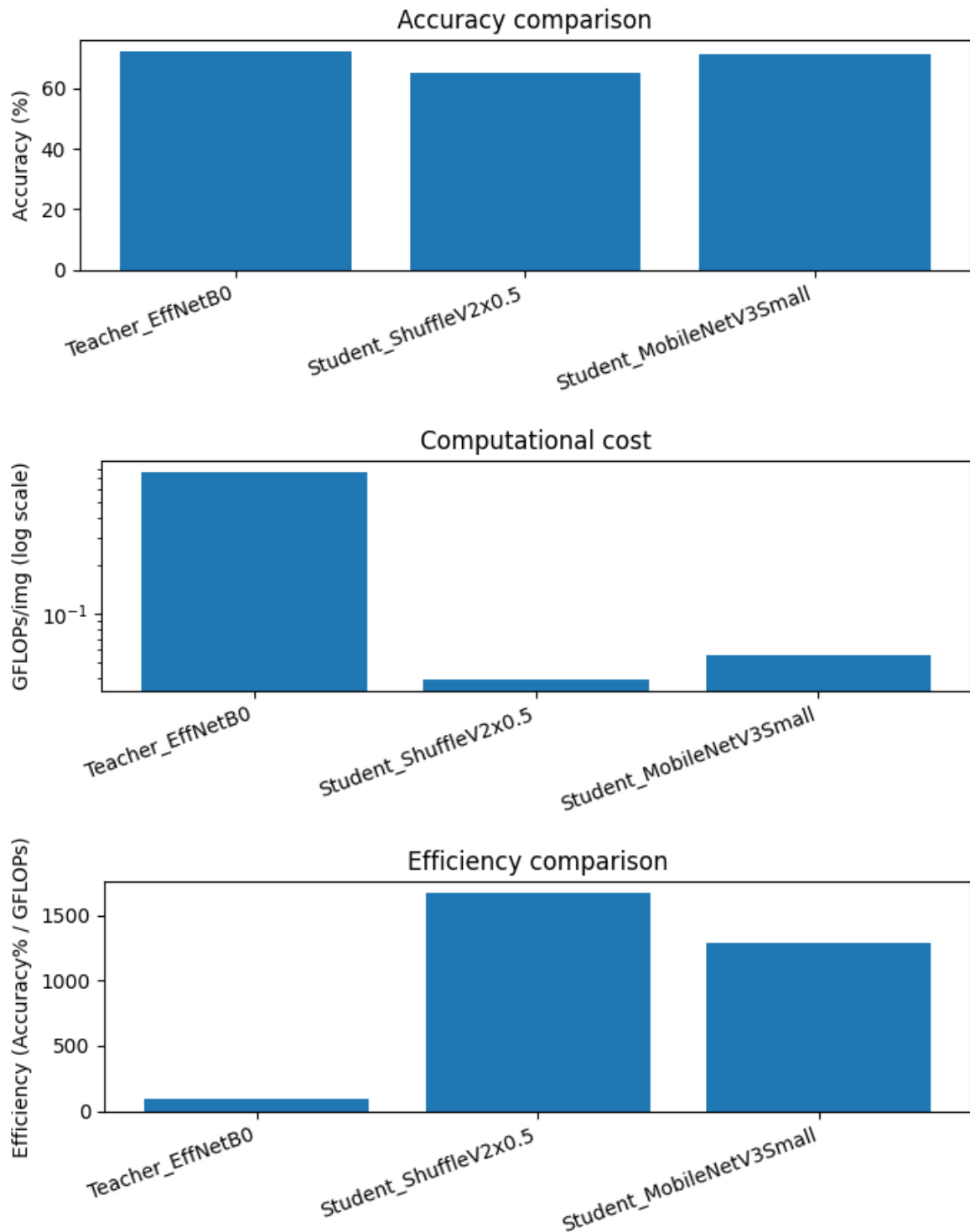
Out[10]:

	batch_size	latency_ms_per_batch	latency_ms_per_image	throughput_imgs_per_sec
0	1	8.146048	8.146048	122.758919
1	8	10.085583	1.260698	793.211479
2	16	6.916046	0.432253	2313.460563
3	32	11.511326	0.359729	2779.870925

Accuracy, FLOPs, and Efficiency Comparison

Out[11]:

	model	accuracy_percent	gflops_per_img	efficiency
0	Teacher_EffNetB0	72.300000	0.769000	94.018205
1	Student_ShuffleV2x0.5	65.200000	0.039000	1671.794872
2	Student_MobileNetV3Small	71.468376	0.055494	1287.867733



This project implements the knowledge distillation to FER2013 facial expression recognition dataset as a small student model (MobileNetV3-Small) trains with a larger teacher model (EfficientNet-B0).

The dataset has been divided into a training, validation and test dataset, and data preprocessing was done by converting to grayscale, augmenting training data, and scaling using ImageNet statistics (Shorten, Khoshgoftaar, 2019). The teacher model (EfficientNet-B0) was also used in training and the student (MobileNetV3-Small) was trained using a combination of cross-entropy loss and soft-target Kullback-Leibler divergence objectives (Hinton et al., 2015).

The student model reached a test accuracy of 71.5%. The results were class specific with the highest F1 attained on the class 3 (0.896) and performance in the other emotions was very good as well (macro F1 = 0.703, weighted F1 = 0.712). Computational efficiency was also calculated. The model consumed 0.055GFLOPs per image only with an efficiency score of 1287.9 (Accuracy% / GFLOPs). These findings show that although MobileNetV3-Small is less accurate than most of its counterparts, it is more efficient in comparison (Howard et al., 2019; Tan & Le, 2019).

In sum, the experiment illustrates that knowledge distillation can be used to deploy highly accurate but very lightweight models of emotion recognition applicable in real-time or limited-resources settings, representing a compromise between accuracy and real-time or low-compute applications.

References

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